TEXORA · LECTURE SHEET

Effects of Moisture

Water inside the fibre: regain and moisture content defined and measured, the sigmoid sorption isotherm and its hysteresis, the heat of sorption, swelling in every direction — and moisture's fingerprints on strength, dimensions, dyeing and electrical behaviour.

TE-3209 · TEXTILE PHYSICS · TOPIC 4 · LEVEL 3 TERM 2

1. Regain and moisture content

Textiles weigh water. Two definitions divide the same water differently, and trade uses one of them:

$$\text{Regain } R = 100 \times \frac{\text{mass of water}}{\text{oven-dry mass}} \quad \cdot \quad \text{Moisture content } M = 100 \times \frac{\text{water}}{\text{total mass}}$$

WORKED EXAMPLE — ONE SAMPLE, BOTH NUMBERS

A conditioned sample weighs 5.00 g; oven drying leaves 4.60 g, so water = 0.40 g. Regain = 0.40/4.60 × 100 = **8.7%**; moisture content = 0.40/5.00 × 100 = **8.0%**. The conversion is $R = 100M/(100 - M)$ and $M = 100R/(100 + R)$ — check: 100 × 8.0/92 = 8.7 ✓. Commerce fixes **commercial (standard) regains** for invoicing — cotton 8.5%, wool 16%, viscose 11%, silk 11%, jute 13.75%, nylon 4.5%, polyester 0.4% — and a bale is paid for at its correct invoice weight: dry weight × (1 + R/100).

2. The sorption isotherm and hysteresis

At fixed temperature, equilibrium regain rises with relative humidity along a sigmoid (S-shaped) curve: slow at first (water binding directly to polar sites), faster through the middle (multilayer sorption), steep near saturation (capillary condensation in fine pores). The path down is not the path up: **hysteresis** leaves a desorbing fibre wetter than an absorbing one at the same RH — by 1–2% regain in wool — because swollen structures and ink-bottle pores empty reluctantly. Consequences: conditioning history matters, test houses specify approach from a defined direction (usually absorption from dry), and a fabric's moisture content in service depends on the weather it has been through, not only the weather now.

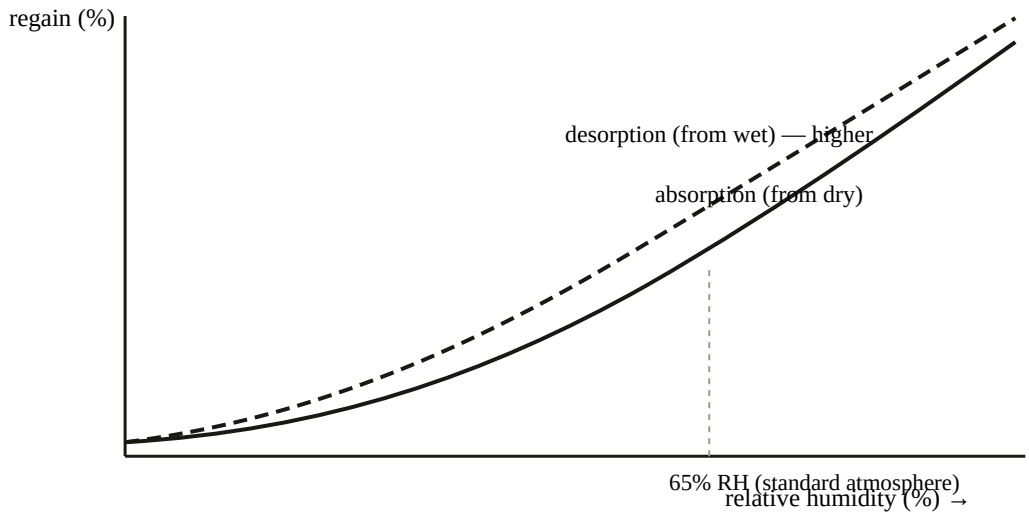


Figure 1 — Sorption isotherm of a hydrophilic fibre (schematic): sigmoid rise, with the desorption branch sitting above absorption — hysteresis. The 65% RH line marks the standard testing atmosphere.

3. Heat of sorption — why wool warms you

Sorption is exothermic: water binding to fibre sites releases energy. The **integral heat of wetting** (dry fibre wetted completely) ranks the fibres' thirst in joules: wool ≈ 113 J/g, viscose ≈ 106, silk ≈ 69, cotton ≈ 47, nylon ≈ 31, polyester ≈ 6. The **differential heat of sorption** is highest for the first water (tightest sites) and falls toward the latent heat of condensation as regain grows. Practical meaning: stepping from cold dry air into humid warmth, a wool garment releases real heat as it sorbs — a kilogram of dry wool taking up 15% regain releases of the order of 10–16 kJ during the change — buffering the wearer against the chill; polyester, sorbing almost nothing, offers no such service.

4. Swelling and the property consequences

Water enters amorphous regions and pushes chains apart — **anisotropically**: fibres swell mainly in diameter (cotton +14–16% diameter, ~+1% length; wool similar pattern) because chains and microfibrils run lengthwise. Area swelling of the order of 20–25% for cellulosics narrows yarn interstices, tightens fabric construction, closes pores (the self-sealing of canvas and the waterproofing of tightly woven cotton), and drives **hygral expansion** and relaxation shrinkage in garments. The property ledger of moisture:

PROPERTY	MOISTURE'S EFFECT
Strength	Cotton +10–20% wet (better load sharing on swelling); wool −15–25%; viscose −40–50%; nylon −10%; polyester nil. (Wet-strength table, TE-3209 Topic 2)
Modulus & recovery	Hydrophilic fibres soften and creep more as regain rises — water is a plasticiser

Dimensions	Swelling → fabric tightness, shrinkage management, hygral expansion in wear
Electrical	Resistance falls orders of magnitude as regain rises (ion paths) — static fades at high RH (Topics 9–10)
Dyeing	Water is the dye's vehicle and gatekeeper: dry fibre barely dyes; controlled wetting-out decides levelness
Thermal comfort	Heat of sorption buffering; high-regain clothing smooths humidity swings against the skin

5. Key points

- Regain (on dry mass) is the trade definition; moisture content (on total mass) converts by $R = 100M/(100 - M)$. Example: 8.7% vs 8.0%.
- The isotherm is sigmoid with hysteresis — desorption sits higher; conditioning history is part of any measurement.
- Heat of wetting (wool 113 J/g down to polyester ≈ 6 J/g) powers moisture buffering in wear.
- Swelling is mostly radial; it tightens fabrics, seals pores and drives shrinkage behaviour.
- Moisture rewrites strength, modulus, electrical resistance and dye uptake — the master variable of Topic 2, quantified.

6. Practice

Answer in your own words; brief model answers follow.

1. A sample has moisture content 12%. Find its regain. A second sample has regain 20%: find its moisture content.
2. A consignment of cotton yarn has oven-dry mass 480 kg. Using the commercial regain of 8.5%, find the correct invoice weight.
3. Why does the desorption curve sit above the absorption curve at the same RH? Give the practical rule for testing laboratories that follows.
4. A wool jumper feels warm when brought into a humid room from cold dry air, though no heater touches it. Explain, naming the quantity involved.
5. Cotton canvas tents waterproof themselves in rain without any coating. Use swelling to explain, and name the property balance that makes cotton (not viscose) right for the job.
6. Predict and explain the direction of change in a cotton fabric's air permeability as RH rises from 30% to 95%.

Answer key

1. R = 100 × 12/(100 − 12) = 13.6%. M = 100 × 20/(100 + 20) = 16.7%.
2. Invoice weight = 480 × 1.085 = 520.8 kg — the buyer pays for 40.8 kg of “water at regain”, the textile trade’s settled convention.
3. Water taken up swells the structure and fills fine pores; on drying, menisci in ink-bottle pores and the relaxed, swollen chain network hold water more strongly, so release lags uptake. Hence laboratories condition specimens toward equilibrium from a specified side — normally absorption after pre-drying — so results are comparable.
4. Sorption is exothermic: as the wool takes up moisture toward its high regain, the differential heat of sorption (integral heat of wetting ≈ 113 J/g from dry) is released inside the garment, giving a measurable warming buffer during the transition.
5. Rain wets the cotton fibres; they swell radially (~15% in diameter), yarn crowns expand and the inter-yarn pores close, so water beads and sheds instead of passing. Viscose would swell too, but its wet strength halves — the tent would fail mechanically; cotton actually strengthens when wet (+10–20%).
6. Permeability falls: rising regain swells fibres and yarns, narrowing the pores through which air passes — the same mechanism that closes tent canvas, measured here as a comfort-relevant property change.

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